

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Regular & Supplementary Summer 2024

Course: B. Tech.

Semester : IV

Branch: Electrical Engineering / Electrical Engineering (Electronics and Power)/

Electrical & Electronics Engg. / Electrical & Power Engineering.

Subject Code & Name: BTEEC402 POWER SYSTEM

Max Marks: 60

Date: 14/06/2024

Duration: 3 Hr.

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

	(Level/CO)	Marks
Q. 1 Solve Any Two of the following.		12
A) Describe the working principle of hydroelectric power Plant. Highlight their advantages, disadvantages, and environmental impacts.	CO1	6
B) With neat diagram explain in detail thermal power plants.	CO3	6
C) Explain in detail Nuclear power plant with neat block diagram of main parts.	CO4	6
Q.2 Solve Any Two of the following.		12
A) Derive the expression for Flux linkages due to a single current carrying conductor.	CO2	6
B) Explain skin and proximity effect in detail.	CO2	6
C) Describe in detail the phenomenon of corona Effect in transmission lines.	CO3	6
Q. 3 Solve Any Two of the following.		12
A) Explain types of Conductor used in transmission line, explain each type in detail.	CO1	6
B) Explain String Efficiency with mathematical expression.	CO4	6
C) Explain types of Insulators used in transmission line, explain with diagram.	CO4	6
Q.4 Solve Any Two of the following.		12
A) Write performance equation of long transmission line.	CO2	6
B) Derive ABCD parameters of medium transmission line (T and π nominal network).	CO3	6
C) A 3-phase, 50-Hz overhead transmission line 100 km long has the following constants :	CO1	6
Resistance/km/phase = 0.1 Ω		
Inductive reactance/km/phase = 0.2 Ω		

Capacitive susceptance/km/phase = 0.04×10^{-4} Siemen

Determine (i) the sending end current (ii) sending end voltage (iii) sending end power factor and (iv) transmission efficiency when supplying a balanced load of 10,000 kW at 66 kV, p.f. 0.8 lagging. Use nominal T method.

Q. 5 Solve Any Two of the following.		12
A) Explain classification of distribution system of power system in detail.	CO2	6
B) Distinguish between overhead distributions systems versus underground distribution system.	CO4	6
C) What are the deferent method to obtain 3 wire DC system in distribution system? Explain with diagram.	CO3	6

***** End *****